

Srikrishnaa Vadivel

Embedded Systems Engineer | FPGA | Robotics | Sensor Logging

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Summary

Embedded systems engineer with field-tested sensor logging, MCU firmware, FPGA logic, robotics, and scientific instrumentation experience. Comfortable from oscilloscope validation and low-level C through host-side visualization and data systems.

Experience

Probe Technologies / Weatherford, Software and Embedded Engineer 2019–2021

- Programmed C8051-class microcontrollers and Lattice FPGA logic for sensor logging in geological well tools.
- Validated embedded systems with oscilloscopes, bench instrumentation, and field deployments.
- Built host-side CUDA/OpenGL software for large acoustic waveform processing and visualization.

Yale University, Embedded Systems Teaching Assistant / PhD Researcher 2021–present

- Served as embedded systems TA while pursuing PhD research in computational materials and scientific computing.
- Used Linux, C/C++, Python, MPI/OpenMP, PETSc/SLEPc, and HPC systems for simulation workflows.

Cornell ECE Robotics Hand Project 2017

- Built an embedded robotic hand project later published in Circuit Cellar.

Portfolio Projects

- **MIPS Processor:** Verilog single-cycle CPU and self-checking Icarus Verilog test bench.
- **Matrix-Multiply Accelerator:** small AI accelerator block with start/done handshake and test bench.
- **ROS 2 TurtleBot Simulation:** robotics simulation plan with trajectory analysis artifacts.

Education

Yale University, PhD Electrical Engineering, 2021–present

Cornell University, MEng ECE, 2017–2018

Cornell University, BA ECE, 2013–2017

Skills

C, C++, Verilog/SystemVerilog, FPGA, MCU firmware, FreeRTOS concepts, SPI, I2C, UART, ADC logging, oscilloscopes, Linux, Python, CUDA, OpenGL, Git.